

GRAVITY CLASSES

"Come Gravity Feel Success"

11th & 12th BOARD
(NEET & JEE)

5th - 10th (All Subject)

NOTES

PHYSICS

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WORK AND ENERGY

Workdone:-

- It states that, "when a force acts on an object and the object, we call it Force has done work on the object".
- In other words, "workdone is defined as product of Force (Vector) [F] and Displacement (Vector) [S] done in the direction of the force".

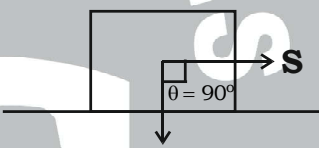
i.e. $W = F \cdot S$ (Nm = J) unit

- Workdone is **Scalar** Quantity.
- Workdone is **Scalar Quantity**.
- **S.I** unit of workdone is **Joule (J)** and **CGS** unit of workdone is **Ergs**.
- Define 1 Joule work:-

$W = F \cdot S$ $Nm = J$

- When a force of 1N is applied on body to displacement it by 1m the workdone is said to be 1 Joule.

$1J = 1N \cdot 1m$

Calculation of Work		
Positive Work	Negative Work	Zero Work
$\theta = 0^\circ$  for ex. $F = 10N$, $S = 5m$ $W = F \times S$ $= 50 J$	$\theta = 180^\circ$ 	 $F = mg$ → Force is nothing but same as 'mg'.
→ If F and S has same direction WD is +ve.	→ If F and S are in opposite direction WD is -ve.	→ If 'F' and S are at right angle with each other WD = 0.
→ Angle between F and S is 0° .	→ Angle between F and S is 180° .	→ Angle between F and S is 90° .
→ It increases its speed (acceleration).	→ It will decreases the speed (Retardation).	→ It will not affect the motion of the object along the displacement.

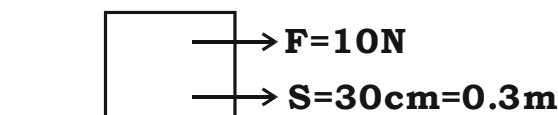
Q. A girl pushes a book by applying a force of 100 N. Find the work done by this force in displacing the book through 30 cm along the direction of the push.

Sol. We know,

$$WD = F \cdot S$$

$$= 10 \times 0.3$$

$WD = +3 J$ (+ve workdone)



Q. A ball of mass 2 kg thrown upwards reaches a maximum height of 7.0 m. Calculate the WD by the force of gravity during this vertical direction.

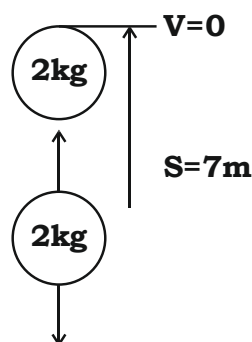
Sol. $F = mg$
 $= 2 \times 10$

$F = 20\text{N}$

Now, $W = F.S$

$= (-20) \times (+7)$ [$\because$ Acceleration due to gravity]

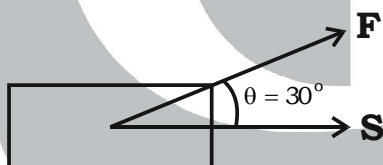
$W = -140\text{J}$ [**-ve** shows **Retardation**]



Q. Why do we pay coolie ?

$W_{D_{coolie}} = 0$ $W_{D_{gravity}} = 0$	$WD = +ve$ $\rightarrow$ Load displaced vertically. i.e. Accelerate	$WD = -ve$ $\rightarrow$ Load displaced downward i.e. Retardation
$W_{D_{coolie}} = 0$ $W_{D_{gravity}} = 0$	$WD = +ve$	$WD = -ve$
	$\rightarrow$ Load displaced vertically. i.e. Accelerate	$\rightarrow$ Load displaced downward i.e. Retardation

● **Force at an angle with the Displacement.**



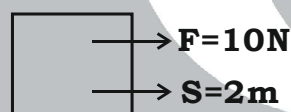
$W.D = F.S.\cos\theta$

Ex. $W.D = F S \cos\theta$

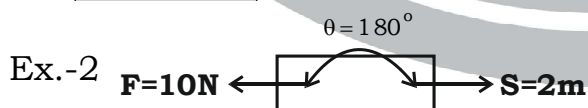
$= 10 \times 2 \cos 0^\circ$

$= 20 \times 1$

$W.D = 20\text{J}$



θ	$\cos\theta$
0°	$\cos 0^\circ = 1$
30°	$\cos 30^\circ = \frac{\sqrt{3}}{2}$
45°	$\cos 45^\circ = \frac{1}{\sqrt{2}}$
60°	$\cos 60^\circ = \frac{1}{2}$
90°	$\cos 90^\circ = 0$
180°	$\cos 180^\circ = -1$

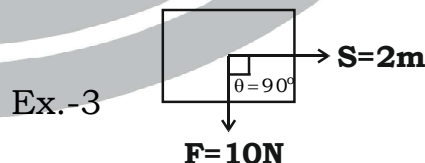


$WD = F S \cos\theta$

$= 10 \times 2 \times \cos 180^\circ$

$= 20 \times -1$

$WD = -20\text{J}$



$WD = F S \cos\theta$

$= 10 \times 2 \times \cos 90^\circ$

$= 10 \times 20 \times 0$

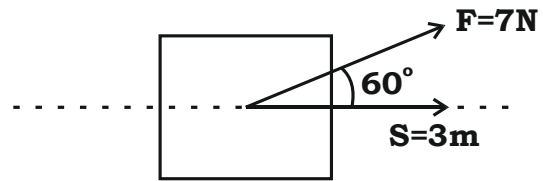
$WD = 0$

Q. A person pulls a block on a horizontal surface by applying a force of 7.0N at an angle of 60° with the horizontal. Find the work done by these force in displacing the block through 3.0m.

Sol. $WD = F S \cos \theta$

$$= 7 \times 3 \times \cos 60^\circ$$

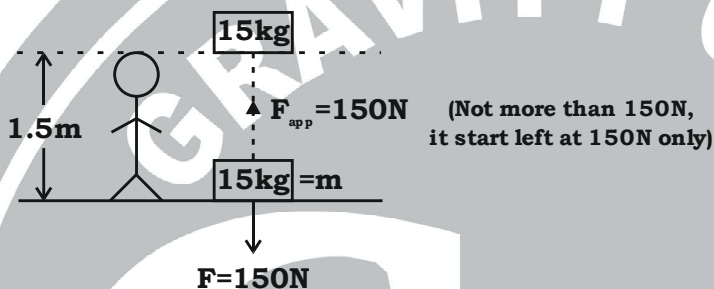
$$= 7 \times 3 \times \frac{1}{2}$$



$$WD = 10.5J$$

Q. A porter lifts a luggage of 15kg from the ground and puts it on his head 1.5m above the ground. Calculate the WD by him on the luggage.

Sol.



$$15\text{ kg}$$

$$\downarrow$$

$$mg$$

$$F = m \times a$$

$$= 15 \times 10 \text{ (} a = 9.8 \approx 10 \text{ m/s}^2 \text{)}$$

$$F = 150N$$

$$\begin{aligned} \text{Now, } WD_{\text{by coolie}} &= F_{app} \times S_{\text{height}} \\ &= 150 \times 1.5 \\ &= 150 \times \frac{1.5}{1} \end{aligned}$$

$$WD = +225J$$

'+ve' sign show displacement in upward direction.

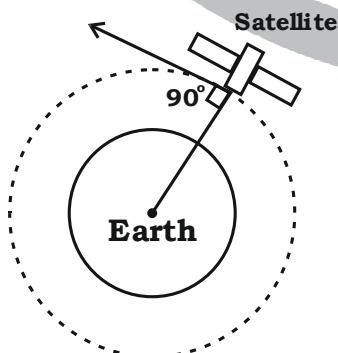
Q. Does the transfer of energy.

$$WD = 0$$

$$\vec{F} \Rightarrow \textcircled{S=0}$$

Molecules and Atom only vibrate but NOT MOVE.

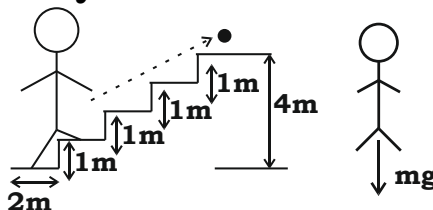
Q. NCERT (Pg-158)



$$WD = 0$$

Q. Find the workdone by the gravity on the man of mass 60 kg while going up the escalator.

Sol. $F = ma$
 $= 60 \times 10$
 $= 600 \text{ N}$



Note:- Displacement Force K along is going on i.e. vertical. (since it is looking inclined displacement).

So, $WD = F.S$
 $= (-600) 4$ [-ve because downwards]

$WD = -2400 \text{ J}$

Q. Find the workdone by all the forces acting on the block and also find the net WD.

(i) $WD_f = F.S$ $= (-5) (+10)$ $WD_f = -50 \text{ J}$	(ii) $F = mg = 5 \times 10 = 50 \text{ N}$ $WD = F S \cos \theta$ $= 50 \times 10 \cos 90^\circ$ $= 50 \times 10 \times 0$ $WD = 0$
(iii) $WD = F.S$ $= 20 \times 10$ $WD = 200 \text{ J}$	(iv) $WD_{\text{net}} = WD_1 + WD_2 + WD_3$ $= -50 + 0 + 200$ $WD_{\text{net}} = 150 \text{ J}$

Sol.

Energy:-

- It states that, "Capability to do workdone".
- In other words, "Energy is required **to do work** and both are interchangably converted".
- Energy is a **Scalar Quantity**.
- **S.I.** unit of Energy is **Joule (J)** and **CGS** unit is '**Ergs**'.
- WD can be changed into numerous form of energy and vice versa.

Kinetic Energy (K.E):-

- It states the "Energy possessed by a body by virtue of its state of motion".
- In other words, "when a body is in motion, it has an Energy which is called **Kinetic Energy** as it derives from a **Greek** word '**Kinein**' which means **motion**".
- If a **body** is at **Rest**, it has **zero K.E**.

→ Formula

$$\frac{1}{2}mv^2$$

Whereas

m = Mass of the body (kg)

v = Velocity of the body (m/s).

Finding Expression of Kinetic Energy

Workdone = Force.displacement

$$W.D = F. S$$

$$W.D = ma. s$$

$$[\because F = ma]$$

$$W.D = m \left(\frac{v^2 - u^2}{2} \right)$$

$$\left[\begin{aligned} \because 2as &= v^2 - u^2 \\ as &= \frac{v^2 - u^2}{2} \end{aligned} \right]$$

$$\boxed{W.D = \frac{mv^2}{2} - \frac{mu^2}{2}} \quad \dots(i)$$

Let the body be initially at rest.

So, i.e. $u = 0$ using eq (i)

$$W = \frac{mv^2}{2} - \frac{m(0)^2}{2}$$

$$W = \frac{mv^2}{2} - 0$$

$$W = \frac{1}{2}mv^2$$

$$\boxed{K.E. = \frac{1}{2}mv^2}$$

Where, K.E. = Kinetic Energy (J)

m = Mass (kg)

v = velocity (m/s)

$$\text{Now, } W = \frac{mv^2}{2} - \frac{mu^2}{2}$$

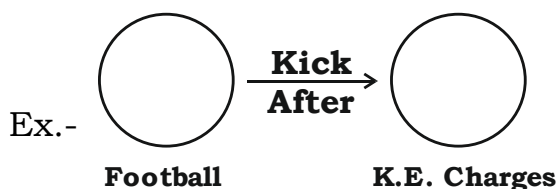
$$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

'**v**' means **Final** Velocity and '**u**' means **Initial** Velocity.

$$W = KE_{\text{final}} - KE_{\text{initial}}$$

$$\boxed{W = \Delta KE} \leftarrow \text{Work Energy Theorem}$$

Workdone = Change in K.E.



Q. How much work is done in changing the velocity of the body of mass 5kg from 10 m/s to 15 m/s ?

Sol. Given $m = 5\text{kg}$, $u = 10\text{ m/s}$, $v = 15\text{m/s}$

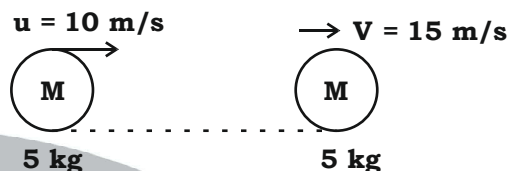
We know, $W = \Delta KE$

$$= K.E_f - K.E_i$$

$$= \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$= \frac{1}{2} \times 5 \times (15)^2 - \frac{1}{2} \times 5 \times (10)^2$$

$$W = 312.55\text{J}$$



Q. Find the WD required to stop a body of mass 10kg travelling at 54 km/hr.

Sol. Given $m = 10\text{ kg}$, $v = 0\text{ m/s}$

We know, $W.D = \Delta K.E$

$$u = 54\text{ km/hr}$$

$$= \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$= 54 \times \frac{5}{18}\text{ m/s}$$

$$= \frac{10}{2}(0^2 - 15^2)$$

$$= 15\text{ m/s}$$

$$W = -1125\text{J} \quad (\text{-ve Retard W.D})$$

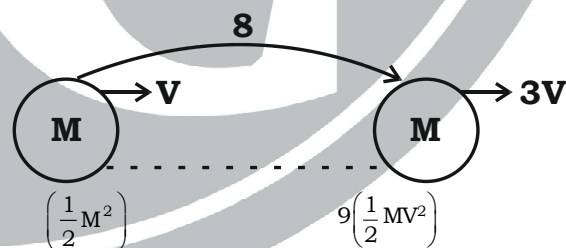
Q. Find the change in K.E. of an object of mass 'M' travelling at 'V' changes its velocity to '3V'. Also find the percentage change in K.E.

Sol. $\Delta K.E. = KE_{2f} - KE_i$

$$= \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$= \frac{1}{2}M(3v)^2 - \frac{1}{2}M(v)^2$$

$$= \frac{1}{2}M9v^2 - \frac{1}{2}Mv^2$$



$$\Delta KE = 8 \times \frac{1}{2}MV^2$$

Note:- Previous K.E. only $\frac{1}{2}mv^2$ after change in K.E. 9 times i.e. $9\left(\frac{1}{2}mv^2\right)$

so it increase by **8 times**.

ii.
$$\% \Delta KE = \frac{KE_f - KE_i}{KE_i} \times 100$$

$$\left[\because \% \text{ change} = \frac{F - I}{I} \times 100 \right]$$

$$= \frac{9 \frac{1}{2} mv^2 - \frac{1}{2} mv^2}{\frac{1}{2} mv^2} \times 100$$

$$= \frac{\frac{1}{2} mv^2 (9 - 1)}{\frac{1}{2} mv^2} \times 100$$

$$= 8 \times 100$$

$$\% \Delta K.E = 800\%$$

#

Potential Energy (P.E):-

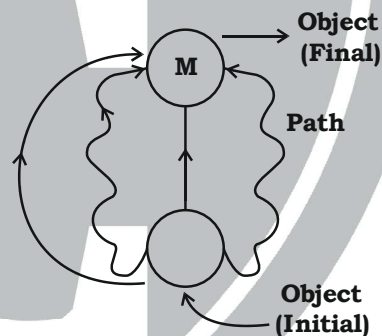
- It states that "**Energy possessed** by a body by virtue of its **position** or **orientation**".
- **By position** "When a body has a **certain position**, it has an Energy which is called **P.E.** due to its position".
- **By orientation** "If a body is **capable** of **storing/holding energy** due to its configuration, the energy stored is called P.E. due to its orientation".
- We **consider Ground** to be at **Zero Level** so potential energy **at ground** is **Zero**.
- It is **independent** of path.
- Formula, $P.E. = mgh$

Where,

m = mass of the body (kg)

g = Acceleration due to gravity (m/s²)

h = Height (m)



Finding the expression of Potential Energy (P.E) Position.

We know,

$$W = \text{Force} \times \text{Displacement}$$

$$W = F_{\text{app}} \times S$$

$$W = ma \times h$$

$$W = mg \times h \quad [F_{\text{app}} = mg, \text{ similar not exactly}]$$

$$\boxed{P.E. = mgh}$$

As, we know

$$(WD = \Delta K.E)$$

So, $\boxed{WD = \Delta P.E}$

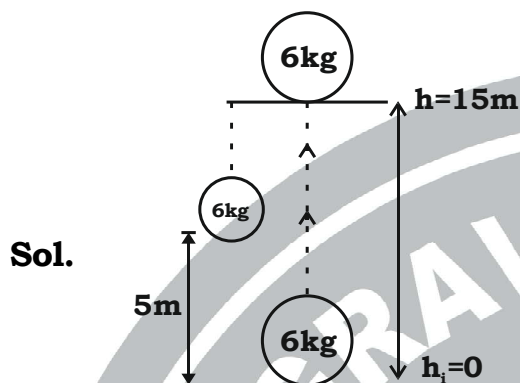
$$W.D = P.E_f - P.E_i$$

$$\boxed{WD = mgh_f - mgh_i}$$

or

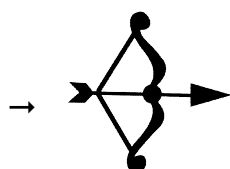
$$\boxed{WD = mgh_2 - mgh_1}$$

Q. Find the W.D. required to raise an object of mass 6 kg to a height of 15 m and then bring down to the height of 5m. Also find the net W.D.

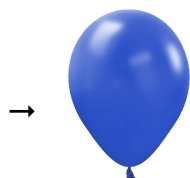


Upward (U) (↑)	Downward (D) (↓)	Total
$WD = \Delta P$ $= P.E_f - P.E_i$ $= mgh_f - mgh_i$ $= 6 \times 10 \times 15 - 6 \times 10 \times 0$ $= 900 - 0$ $\boxed{WD_u = 900 \text{ J}}$	$WD = F.S$ $= mg.s$ $= 6 \times 10 \times 10$ $[\because S = 15 - 5]$ $= 6 \times 10 \times -10$ $= (-ve \text{ downward})$ $= 6 \times 10 \times -10$ $\boxed{WD_d = -600 \text{ J}}$	$WD_N = WD_u + WD_d$ $= 900 + (-600)$ $\boxed{WD_N = 300 \text{ J}}$
	$WD = \Delta P.E$ $= P.E_f - P.E_i$ $= mgh_f - mgh_i$ $= 6 \times 10 \times 5 - 6 \times 10 \times 15$ $= 6 \times 10 (5 - 15)$ $= 60 (-10)$ $\boxed{WD_d = -600 \text{ J}}$	$WD = \Delta P.E$ $= P.E_f - P.E_i$ $= mgh_f - mgh_i$ $= 6 \times 10 \times 5 - 0$ $[\because h_i = 0 \text{ bcz. Ground Level}]$ $= 6 \times 10 \times 5 - 0$ $\boxed{WD_N = 300 \text{ J}}$

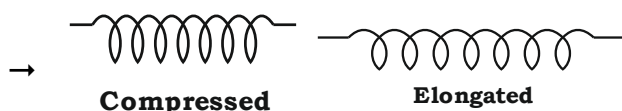
Understanding P.E. (Configuration):-



in the bow.



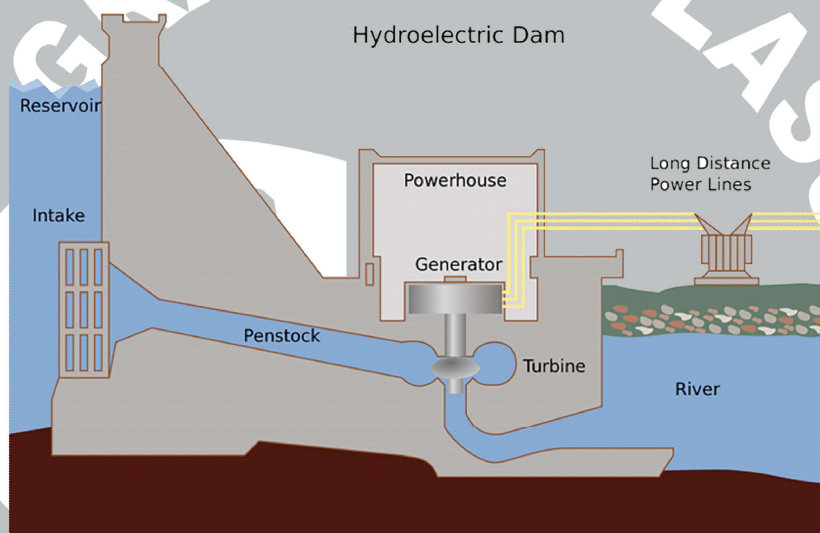
An inflated **balloon** stores a potential energy.



A compressed or elongated **spring** has potential energy stored in it.

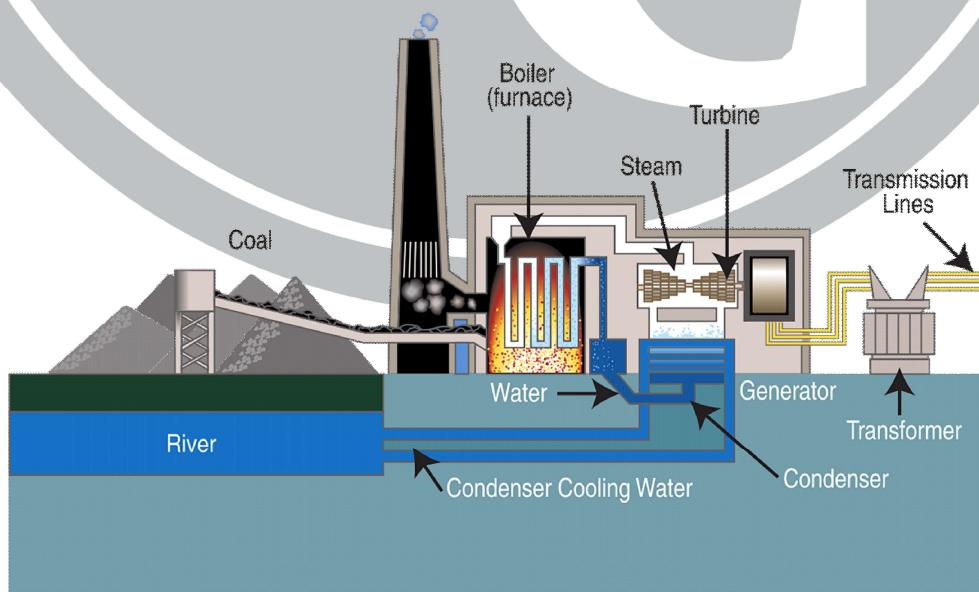
● **Law of Conservation of Energy:-**

- It states that "Energy can neither be created nor be destroyed, it can only changes from one form to another form".
- In other words "**total energy** in the universe is **constant** and divided in many forms which are interchangeably connected with workdone".
- ex.- (i) Riding a bicycle, (ii) Eating Food, (iii) Burning of Coal (Carbon Present).
- **Electric Bulb:-** Electrical Energy → Light Energy → Heat Energy.
- **Electric Heater:-** Electrical Energy → Heat Energy (Thermal).
- **Electric Fan:-** Electrical Energy → Rotational K.E. + Heat Energy.
- **Hydel Power Plant:-**



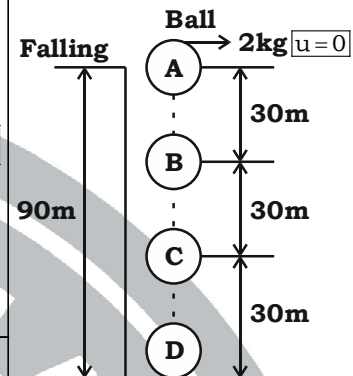
Note:- P.E. → K.E. → R.K.E. → Electrical Energy.

- **Thermal Power Plant:-** Chemical Energy → Thermal Energy (Heat) → K.E. → R.K.E. (Electrical Energy).



● Proof of Conservation of Mechanical Energy (Analytical Method)

	P.E.(mgh)	K.E($\frac{1}{2}mv^2$)	T.E.= Const. (K.E+P.E)
A	$m \times g \times h$ $2 \times 10 \times 90 = 1800 \text{ J}$	$\frac{1}{2} \times m \times v^2$ $\frac{1}{2} \times 10 \times 0 = 0 \text{ J}$	$1800 + 0 = 1800 \text{ J}$
B	$2 \times 10 \times 60 = 1200 \text{ J}$	$\frac{1}{2} \times 2 \times v^2$ For v^2 , $2as = v^2 - u^2$ $2 \times 10 \times 30 = v^2 - 0^2$ $600 = v^2$ $\frac{1}{2} \times 2 \times 600 = 600 \text{ J}$	$1200 + 600 = 1800 \text{ J}$
C	$2 \times 10 \times 30 = 600 \text{ J}$	$\frac{1}{2} \times m \times v^2$ For V^2 , $2as = v^2 - u^2$ $2 \times 10 \times 60 = v^2 - 0^2$ $1200 = V^2$ Now, $\frac{1}{2} \times 2 \times 1200 = 1200 \text{ J}$	$600 + 1200 = 1800 \text{ J}$
D	$2 \times 2 \times 0 = 0 \text{ J}$	$\frac{1}{2} \times 2 \times (2as)$ $2 \times 10 \times 90 = 1800 \text{ J}$	$0 + 1800 = 1800 \text{ J}$



Power:-

- It states that "**rate of doing work** or **rate of transfer of energy**".
- In other words "kisi kaam ko kitni jaldi kiya jaa raha hai is known as **Power**".
- Power is a **Scalar Quantity**.
- **S.I.** unit of Power is **Watt (W)** and another Unit is **Horsepower (1 Hp = 746 W)**.

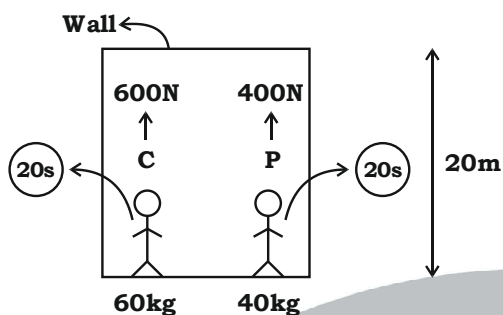
→ Formula, $\text{Power}(P) = \frac{\text{Workdone}}{\text{Time}}$

$$P = \frac{W}{T}$$

$$P = \frac{\text{Change in Energy}}{\text{Time}}$$

$$P = \frac{E}{T}$$

Q. "Chinky (60 kg) and Pinky (40 kg) climb a wall of height 20m in 20s and 10s respectively. Who is more powerful ?



Sol.

$$F = ma \\ = 60 \times 10 \\ = 600\text{N}$$

$$F = ma \\ = 40 \times 10 \\ = 400\text{N}$$

Chinky	Pinky
$m = 60 \text{ kg}$	$m = 40 \text{ kg}$
$h = s = 20 \text{ m}$	$h = s = 20 \text{ m}$
$F = ma$	$F = ma$
$= 60 \times 10 = 600 \text{ N}$	$= 40 \times 10 = 400 \text{ N}$
$t = 20 \text{ s}$	$t = 10 \text{ s}$
So, $P = \frac{W}{T}$	So, $P = \frac{W}{T}$
$= \frac{F \times S}{T}$	$= \frac{F \times S}{T}$
$= \frac{600 \times 20}{20}$	$= \frac{400 \times 20}{10}$
P = 600 Watt ↓	P = 800 Watt ↑

* Pinky is more powerful.

Commercial (Money) unit of Energy.

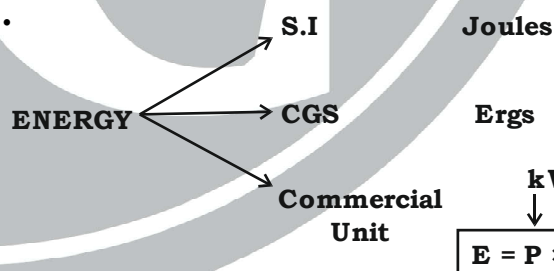
"1 kWh = 1 kilo Watt hour = 1 unit".

$$P = \frac{W}{T} = \frac{E}{T}$$

$$P = \frac{E}{T}$$

$$E = P \times T$$

$$[(P - \text{Watt}) \times (\text{Sec})]$$



Kilowatt × Hour (kWh)

Q. A set of 4 Bulbs of power rating 40 W each are being run for 10 hours each day in the month of September. Find

(i) Total Power Expended.

(ii) Total Energy Consumption.

(iii) Total cost of Electricity bill if rate being Rs. 3 /unit.

Sol. (i) 1 bulb = 40 W

So, 4 bulb = $40 \times 4 \Rightarrow 160$

Power of 4 bulb = **160 W**

(ii) Energy consumption (E)

So, $E = P \times T$

= $160 \text{ W} \times 10 \text{ Hr.}$

= $\frac{160 \text{ kW}}{1000} \times 10 \text{ Hr}$

= $\frac{160 \times 10}{1000} \text{ kWh}$

(for 1 day) $\leftarrow E = 1.6 \text{ kWh.}$

* A/c September (30 days).

$E = 1.6 \times 30 \text{ kWh}$

E = 48 kWh = 48 unit

(iii) Cost of 1 unit = 3 Rs.

48 unit = 48×3

= **144 Rs**



GRAVITY CLASSES

"Come Gravity Feel Success"

11th - 12th
NEET, IIT/JEE

5 - 10th
ICSE & CBSE BOARD



MD REHAN RAZA
LITERA VALLEY SCHOOL

94%

Xth (CBSE)
2025
RESULT

2ND
RANK
IN SCHOOL

HIBA AHMAD
MOUNT ASSISI SCHOOL

94%



ASAD HAQUE
DELHI PUBLIC SCHOOL

87%



ALVINA TANVEER
BISHOP SCOTT GIRLS SCHOOL

88%

1st
RANK
IN SCHOOL

MD SHALIN IRSHAD
BLUE PEARL HIGH SCHOOL

87%



97%



SHADMAN ALI

93%



KASHAF EJAZ

91.4%



ALIYA AFREEN

TOPPERS
2024

ER. AMIR SIR / ER. ASAD SIR  **7004166363, 7717752909**